MILITARY SPECIFICATION

ELECTRONIC UNIT: 11743131

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Electronic Unit: 11743131 which is a unit of the Laser Rangefinder, 11743101.

2. APPLICABLE DOCUMENTS

2. Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

FSC 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARRADCOM

11743131	Electronic Unit

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750439	Adapter - Electronic Unit
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
11821471	Adapter - Handling Fixture, Altitude
11821520	Automatic Test Assembly Central Processor Unit
11821748	Immersion Leak, Purge and Fill

Packaging Data Sheet

P11743131	Packaging of Electronic Unit: 11743131

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Electronic Unit, herein referred to as the unit, shall be manufactured in accordance with Drawing 11743131 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The unit shall provide the following functions:

	Designation
a.	System power supply and control
b.	Laser pulse forming
c.	Reply gating and selection
d.	Range counting
e.	Computer input line driving
f.	Malfunction detection
g.	Emergency power control

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° ± 18°F
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 − 3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirments specified herein under standard ambient conditions of 3.1.3.

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3.3.1 Circuit continuity and isolation.

3.3.1.1 Continuity. With no power or signals applied, direct continuity which is resistance not greater than 2.0 ohm, shall exist between pins listed in table II.

3.3.1.2 Signal return isolation. With no power applied, the resistance between J6-L and J6-S shall be not less than one (1) megohm. (See 4.6.2.1)

TABLE I. Loads, power and signals.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connect between the following pins:
1.1	Fixed resistor	130 ohm ± 5 percent, 2 watts	J3-g, h, i, j, k, individually and J3-L (14 Vdc)
1.2	Fixed resistor	100 kilohms ± 1 percent, 1/2 watt	J2-u and J2-v unless otherwise specified
1.3	Fixed resistor	32.4 kilohms ± 1 percent, 1/2 watt	J2-u and J2-v when specified
1.4	Fixed resistor	66.5 kilohms ± 1 percent, 1/2 watt	J7-A and J7-J when specified
1.5	Fixed resistor	240 ohm ± 5 percent, 2 watts	J3-b, c, d, m, n, p individually and J3-L (14 Vdc)
1.6	Fixed resistor	2 kilohms ± 5 percent, 1/4 watt	Connected as specified in Table V
1.7	Fixed resistor	1 kilohms ± 5 percent, 1/4 watt	J2-R and J2-U
1.8	Fixed resistor	470 ohms ± 5 percent, 1/2 watt	J3-E, F, G, H, J, and J3-M (rtn)

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TABLE I. Loads, power and signals. - Continued

Item	Conditions	Characteristics	Connections
1.9	Fixed resistor	33 Kilohms ± 5 percent, 1/2 watt	J6-D and J6-L
1.10	Fixed resistor	2 kilohms ± 5 percent, 1/4 watt	Connected as specified in Table IX
1.11	Fixed resistor	910 ohms ± 5 percent, 1/4 watt	J3-P and J3-K (5 Vdc)
1.12	Fixed resistor	1 kilohm ± 5 percent, 2 watts	J2-S and J2-V
1.13	Fixed resistor	56 ohm ± 5 percent, 10 watts	J2-T and J2-V
1.14	Fixed resistor	390 kilohm ± 5 percent, 2 watts	J2-W and J2-V
1.15	Fixed resistor	56 ohm ± 5 percent, 1 watt	J2-X and J2-V
1.16	Fixed resistor	3 megohm ± 5 percent, 2500 V minimum	J2-s and J2-p
2.0	Power Sources		Applied between the following pins unless otherwise specified:
2.1	Tank power	24 ± 6.0 Vdc Ripple: 200 mV p-p max	J4-A (+) and J4-B (+) with negative returns to J4-C and J4-D
2.2	Computer on power	24 ± 6.0 Vdc Ripple: 200 mV p-p max	J6-A (+) with return to J6-L (± 15 V rtn comp)
2.3	15 V computer	15.0 ± 0.7 Vdc Ripple: 25 mV p-p max	J6-J (+) with return to J6-L (± 15 V rtn comp)
2.4	-15 V computer	-15.0 ± 0.7 Vdc Ripple: 25 mV p-p max	J6-K (-) with return to J6-L (± 15 V rtn comp)

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TABLE I. Loads, power and signals. - Continued

Item	Conditions	Characteristics	Connections
2.5	AC Ref. 1 OR DC Ref.	5.000 ± 0.002 Vac 1/ 400 ± 5 Hz 5.000 ± 0.002 Vdc 1/ Ripple: 100 micro V p-p max	J6-M with return to J6-L (± 15 V rtn comp)
3.0	Signal Sources		
3.1	Digital Type A	Logic one = 4.0 1 Vdc Source impendance: 2 kilohms ± 5 percent Logic zero = 0.2 ± 0.2 Vdc	Applied between input pins as specified with return to J2-V
3.2	Digital Type B	See figure 2	See figure 1

1/ The tolerance on the Ref. voltages is increased to ± 0.05 V when ratio measurement method is used.

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TABLE II. Continuity requirements.

Item	From 1/	To 1/	Item	From 1/	To 1/	Item	From 1/	To 1/
1	J2-A	J6-N	15	J2-T	J7-J	29	J2-X	J7-L
2	J2-B	J6-M	16	J2-U	J7-K	30	J2-b	J7-D
3	J2-B	J7-X	17	J2-U	J3-K	31	J2-e	J7-P
4	J2-C	J6-P	18	J2-V	J7-N	32	J2-q	J7-C
5	J2-D	J6-R	19	J2-V	J3-M	33	J2-AA	J7-W
6	J2-E	J6-S	20	J2-V	J4-C	34	J2-AA	J6-Y
7	J2-G	J6-U	21	J2-V	J4-D	35	J2-FF	J3-EE
8	J2-H	J6-V	22	J2-V	J2-Y	36	J2-GG	J2-HH
9	J2-J	J6-W	23	J2-V	J2-p	37	J3-GG	J3-HH
10	J2-K	J6-X	24	J2-V	J3-s	38	J4-A	J4-B
11	J2-L	J6-a	25	J2-V	J6-L	39	J6-B	J7-R
12	J2-M	J6-K	26	J2-V	J6-Z	40	J6-D	J7-Y
13	J2-N	J6-J	27	J2-V	J7-E	41	J6-h	J6-j
14	J2-S	J7-Z	28	J2-W	J7-H	42	J3-L	J7-S
						43	J3-P	J3-M

1/ For complete connector designation, add "A31" in front in listing.

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3.3.2 Power interlock.

3.3.2.1 Card cage cover interlock. With no power applied and card cage cover removed, the resistance between J3-M and J3-FF shall be not less than one megohm. With card cage cover installed, direct continuity, less than or equal to 1.0 ohm, shall exist between J3-M and J3-FF. The resistance between J3-FF and ground shall be not greater than 1.0 ohm.

3.3.2.2 System interlock. With no power, direct continuity shall exist between J2-FF and J3-EE and the resistance between J3-EE and ground shall be not less than one megohm.

3.3.2.3 Emergency power interlock. With EMER POWER S1 switch set to the ON position and with application of 28 V tank power (Item 2.1 of table I), connection of pin J1-D to chassis ground shall activate the low voltage power supply (LVPS) and the output voltage shall be as specified in table III.

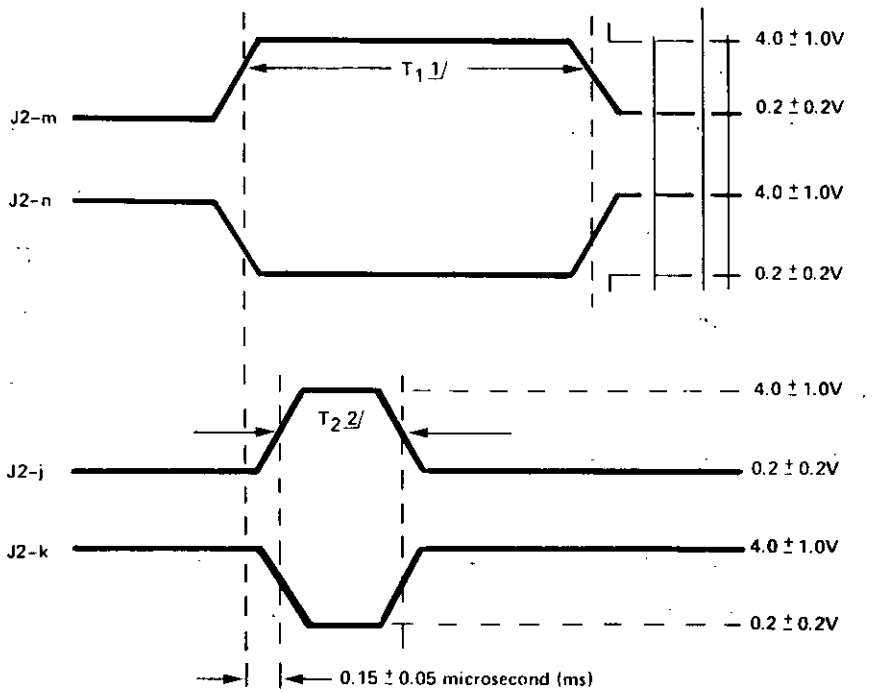


FIGURE 1. Digital type B signal input waveform.

1/ T_1 = 1.0 ± 0.2 ms at 50% amplitude.

2/ T_2 = 0.5 ± 0.1 ms at 50% amplitude.

3/ t_r (rise time) 100 nanoseconds or less (10 to 90% amplitude).

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TABLE III. Primary power application response.

Item	Output Pins	Output Voltages 1/, 2/
1	J2-s	-1900 ± 200 Vdc
2	J2-X	-5.0 ± 0.8 Vdc
3	J2-U	5.2 ± 0.5 Vdc
4	J3-L	14.0 ± 2.0 Vdc
5	J2-T	15.2 ± 0.8 Vdc
6	J2-W	31.0 ± 3.0 Vdc
7	J7-M	24.0 ± 6.0 Vdc
8	J2-S	23.0 ± 6.0 Vdc

1/ Item 1 output voltage referenced to pin J2-p.
2/ Items 2 through 8 output voltages referenced to pin J2-V.

3.3.2.4 LVPS interlock. With LVPS activated, performance of any one of the following operations shall deactivate the LVPS and output voltages at all the output pins listed in table III shall be 0.5 ± 0.5 Vdc.

a. Removal of pulse forming network (PFN).

b. Removal of connection between J1-D and chassis ground. (See 4.6.2.2)

3.3.3 Remote power-on. With pin J1-D connected to chassis ground, tank power (item 2.1 of table I) applied, computer on power (item 2.2 of table I) applied, and with EMER POWER S1 switch in the OFF position, the voltage between pins J7-M (+) and J2-V (-) shall be 24.0 ± 6.0 Vdc. (See 4.6.2.3)

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3.3.4 Initial turn-on conditions and battle range.

3.3.4.1 Initial turn-on. With J1-D connected to chassis ground and a logical zero (item 3.1, table I) applied between J3-W and J3-M (rtn) and with the application of the power sources (items 2.1 and 2.2, table I), the following conditions shall exist:

a. The voltage waveform as specified in figure 2 shall be present between J3-g and J3-M (rtn) within 4 seconds after application of tank power.

b. The voltage between J6-E and J6-L (rtn) shall be 4.5 ± 1.2 Vdc.

c. The analog voltage output between J6-D and J6-L (rtn) shall be 1.2768 ± 0.0040 Vrms (1.2768 ± 0.0040 Vdc when DC Ref. voltage is used) or the ratio of the analog voltage output to the corresponding Ref. voltage (item 2.5, table I) shall be 0.25536 ± 0.00080.

d. The voltage between J3-h and J3-M (rtn) shall be 0.4 ± 0.4 Vdc.

e. The voltage between J3-k and J3-M (rtn) shall be 0.4 ± 0.4 Vdc.

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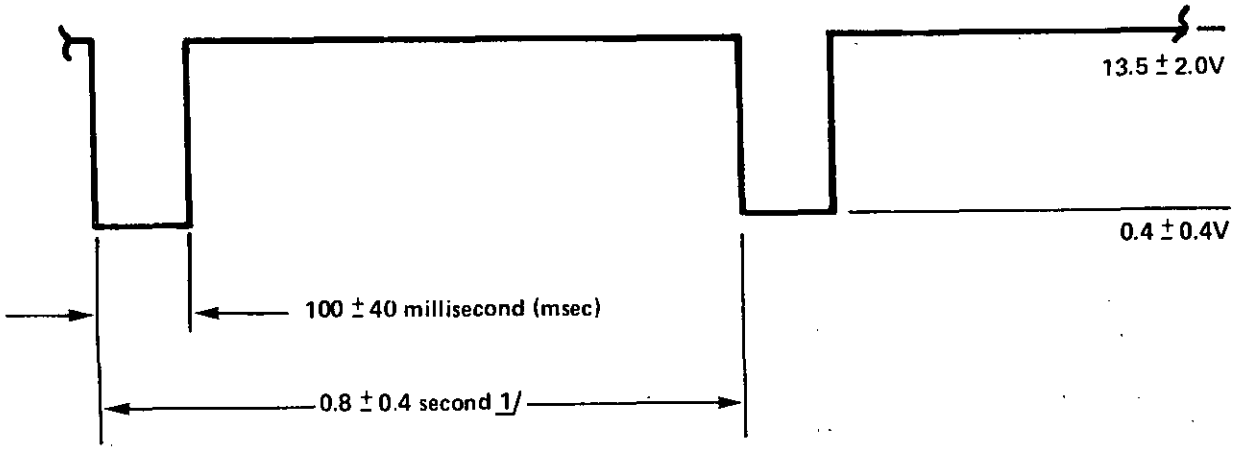


FIGURE 2. Range light return signal waveform.

1/ Shall be 0.8 ± 0.6 second for the temperature extremes of 3.5.3.

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3.3.4.2 Auto range select. With the conditions of 3.3.4.1 established, the application of a logical zero (item 3.1, table I) between J6-F and J6-L (rtn) shall result in a voltage between J3-g and J3-M (rtn) of 0.4 ± 0.4 Vdc. (See 4.6.2.4)

3.3.5 Pulse forming network (PFN) current sensing. With the PFN voltage set at 1200 ± 15 V between J1-A and J1-B, the current between J7-A and J7-N shall be 75 ± 2.5 microamperes. (See 4.6.2.5)

3.3.6 PFN voltage adjustment and programming.

3.3.6.1 PFN voltage adjustment. The PFN voltage between J1-A and J1-B shall be adjustable over a range of 1050 Vdc to 1300 Vdc with a 100 kilohm load between J2-u and J2-v (item 1.2 of table I).

3.3.6.2 PFN voltage programming. With the PFN voltage between pins J1-A and J1-B at 1200 ± 15 V, changing the load between J2-u and J2-v from 100 kilohms (item 1.2 of table I) to 32.4 kilohms (item 1.3 of table I) shall cause the PFN voltage to increase to 1345 ± 40 V. (See 4.6.2.6)

3.3.7 Four hundred volt supply. With the PFN voltage between J1-A and J1-B at 1200 ± 200 V, the voltage between J2-g and J2-V shall be 400 ± 40 V. (See 4.6.2.7)

3.3.8 Fire and Malf 4 signals. With the PFN voltage between J1-A and J1-B (rtn) adjusted to 1200 ± 200 Vdc, the input signal (item 3.1, table I) established for input pins as listed in table IV for each item listed therein and with the table IV inputs established in the order of Items 1 through 7, the corresponding order of output signals shall be as specified therein. With a 66.5 kilohm resistor connected as listed in item 1.4 of table I, the CLRC readout shall display 0004. (See 4.6.2.8)

3.3.9 Malf 2 signal.

a. With J2-d connected to J2-V and a logical zero (item 3.1, table I) applied between J3-R and J3-M (rtn), application for not less than one second and not greater than two seconds of a logical one (item 3.1, table I) between J2-c and J2-V (rtn) shall result in an output voltage between J3-p and J3-M (rtn) of 1.8 ± 0.8 Vdc. The CLRC readouts shall display 0002.

b. The application of a logical zero (item 3.1, table I) between J3-U and J3-M (rtn) after the conditions of 3.3.9.a above have been established shall change the output voltage between J3-p and J3-M (rtn) to 13.5 ± 2.0 Vdc and the readouts to 0000.

c. With the conditions of 3.3.9.a above established, application of the digital type-B signals (item 3.2, table I) shall result in an output voltage between J3-p and J3-M (rtn) of 13.5 ± 2.0 Vdc and the read outs shall display 000 or 9995. (See 4.6.2.9)

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TABLE IV. FIRE and MALF 4 signal outputs.

Item	Input Conditions at A31 Connector 1/, 2/								Output 1/, 4/		Timing
	J3-R	J2-d	J2-c	J3-U	J3-W	J3-AA	J2-P	J6-B	J2-R	J3-B, J3-F	CLRC
1	1	0	1	0	0 3/	OC	1	NA	High		0000
2	1	0	1	1	OC	0 3/	1	NA	High	5/	0004
3	1	0	1	0	0 3/	OC	1	NA	High		0000
4	1	1	0	1	OC	0 3/	1	NA	Low		0000
5	1	1	0	1	0 3/	OC	0 3/	NA	High		0000
6	1	1	0	1	0 3/	OC	1	1	Low		0000
7	0	1	0	1	0	OC	1	NA	High		0000

1/ Input and output signal condition voltages referenced to J2-V.
2/ Input signal legend: 1 = logical one, digital type - A signal
0 = logical zero, digital type - A signal
NA = not applicable
OC = open circuit
3/ Signal applied for not less than one second and not greater than two seconds.
4/ Output condition legend: High = 4.5 ± 1.2 V Low = 0.2 ± 0.2 V
5/ The data signal and the anode signal for the specified pins are synchronous per figure 3.

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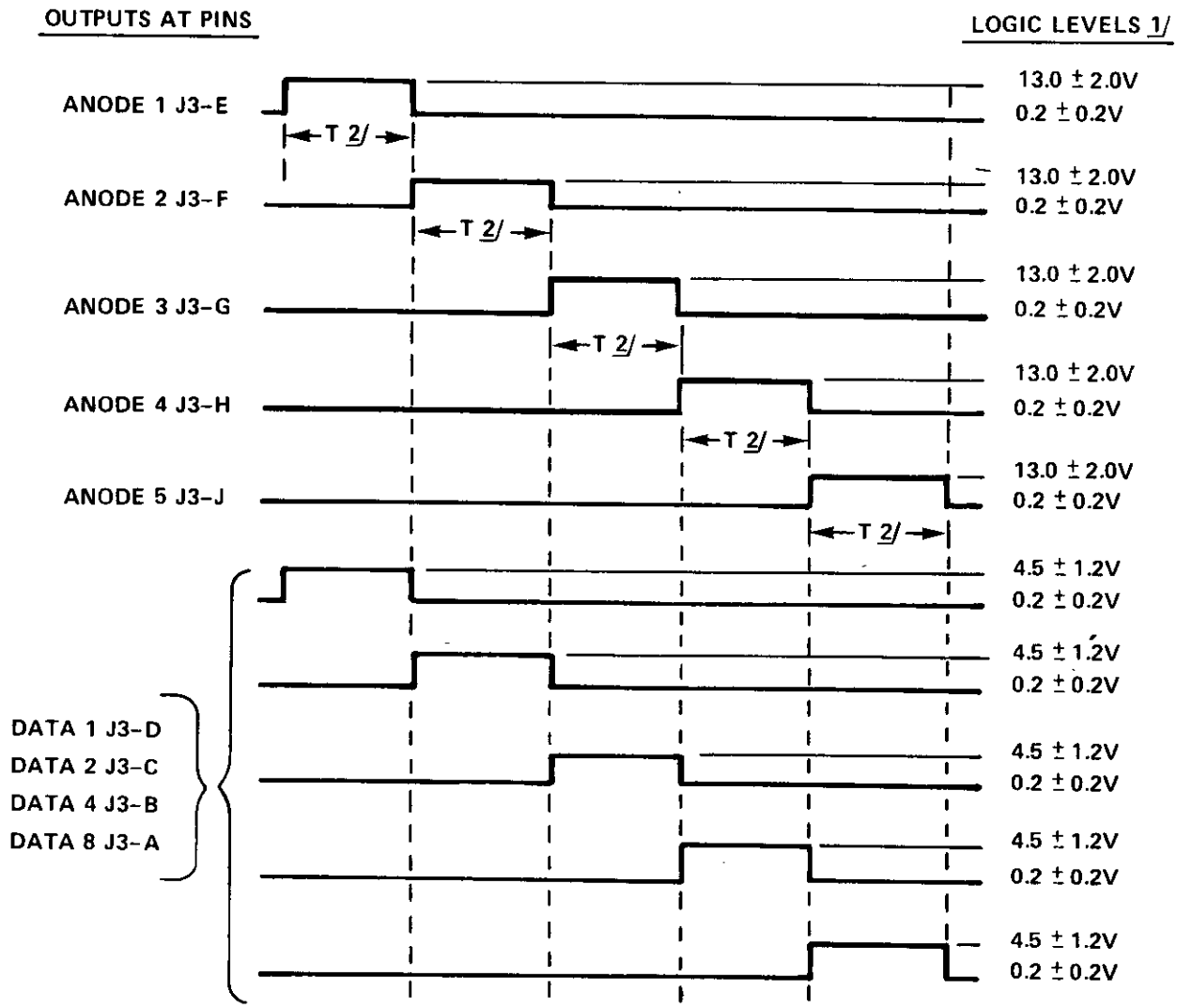


FIGURE 3. Anode and data output waveform.

1/ Voltages referenced to J2-V (rtn).
2/ T = 3.2 ± 0.9 msec.

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3.3.10 Range readout. With Counters Circuit Card Assembly (PN 10559305) removed from the unit and with connected as specified, under the input conditions listed in table V, the output shall be as specified in table V and figure 4. (See 4.6.2.10)

3.3.11 Switch and light control. With each of the output pins of table VI connected to 13.5 ± 2.0 Vdc through loads per items 1.1 and 1.5 of table I and with a logical 0 digital type - A signal (item 3.1 of table I) applied as specified, the unit shall provide outputs in accordance with table VI. With a logical 0 signal applied between J3-r and J2-V the output at J3-g, m, and n, in addition to the output pins listed in table VI shall be 0.4 ± 0.4 Vdc and the output at J3-p shall be 1.8 ± 0.8 Vdc. (See 4.6.2.11)

3.3.12 Blanking. Application of logical zero digital type - A signal (item 3.1 of table I) as specified and in the order listed in 3.3.12.a through 3.3.12.c below, shall provide an output waveform between J3-q and J2-V as shown in figure 5. (See 4.6.2.12)

a. Continuously between J3-U and J2-V.

b. For not less than one second and not greater than two seconds between J3-W and J2-V.

c. For not less than one second and not greater than two seconds between J3-AA and J2-V.

3.3.13 $\overline{\text{MANUAL RESET}}$. With logical zero digital type - A signal (item 3.1 of table I) applied between J3-U and J2-V (ground), the outputs between J7-P and J2-V, and between J2-e and J2-V shall be 0.2 ± 0.2 V. With J3-U in the open condition, the outputs at J7-P and J2-e, with respect to J2-V, shall be 4.5 ± 1.2 Vdc. (See 4.6.2.13)

3.3.14 Test access outputs. With logical zero digital type - A signal (item 3.1 of table I) applied between J3-R and J2-V, and with digital type - A signals applied to the input pins listed in table VII, the test access outputs shall be in accordance with table VII. (See 4.6.2.14)

3.3.15 Target select, $\overline{\text{reset}}$, $\overline{\text{RBI}}$ (ripple blanking input), $\overline{\text{feed}}$, and $\overline{9995}$. With J2-h connected to J2-j and J2-k connected to J2-i, the connected to the unit as specified, a logical zero digital type - A signal (item 3.1 of table I) applied between J3-R and J2-V, and with the input conditions as established in the order of items 1 through 15, the corresponding ordered outputs shall be as specified in table VIII. (See 4.6.2.15)

3.3.16 Battle range logic. With a logical zero digital type - A signal (item 3.1 of table I) applied between J3-X and J3-M, the output between J6-E and J6-L shall be 4.5 ± 1.2 Vdc. With conditions of items 1 and 2 of table VIII established in the item numbered order, the output between J6-E and J6-L (rtn) shall be 0.2 ± 0.2 Vdc. (See 4.6.2.16)

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TABLE V. Output range.

Item	Input Conditions 2/ A1XA5 Pins 1	2	3	4	5	6	7	8	9	10	11	12	13	Output Voltage/Timing J3-D J3-G	J3-C J3-H	J3-D J3-F	J3-A J3-J	J3-B J3-F	CLRC Output Range
1	G	+	+	+	+	+	+	+	+	+	+	+	+				1/		8000
2	+	G	+	+	+	+	+	+	+	+	+	+	+						4000
3	+	+	G	+	+	+	+	+	+	+	+	+	+						2000
4	+	+	+	G	+	+	+	+	+	+	+	+	+						1000
5	+	+	+	+	G	+	+	+	+	+	+	+	+						800
6	+	+	+	+	+	G	+	+	+	+	+	+	+						400
7	+	+	+	+	+	+	G	+	+	+	+	+	+		1/				200
8	+	+	+	+	+	+	+	G	+	+	+	+	+						100
9	+	+	+	+	+	+	+	+	G	+	+	+	+						80
10	+	+	+	+	+	+	+	+	+	G	+	+	+						40
11	+	+	+	+	+	+	+	+	+	+	G	+	+						20
12	+	+	+	+	+	+	+	+	+	+	+	G	+	1/					10
13	+	+	+	+	+	+	+	+	+	+	+	+	G			1/		1/	5

1/ The data signal and the anode signal for the specified pins are sychronous per figure 4.
2/ Legend: + indicates pin connected to 5V through a 2 kilohm resistor (item 1.6 of table I).
G indicates pin connected to return J2-V.

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TABLE VI. Switch and light control performance.

Item	Input Conditions 1/ 2/ J3 Pins						Output Condition 1/ 2/ J3 Pins					
	R	S	T	Y	Z	a	b	c	d	i	j	k
1	0	OC	OC	OC	OC	OC	ON	OFF	OFF	OFF	OFF	ON
2	OC	0	OC	OC	OC	OC	OFF	ON	OFF	OFF	OFF	ON
3	OC	OC	0	OC	OC	OC	OFF	OFF	ON	OFF	OFF	ON
4												OFF
5	OC	OC	OC	0	OC	OC	OFF	OFF	OFF	ON	OFF	OFF
6	OC	OC	OC	OC	0	OC	OFF	OFF	OFF	OFF	ON	ON
7	OC	OC	OC	OC	OC	0	OFF	OFF	OFF	OFF	OFF	

1/ Input and output signal voltages referenced to pin J2-V.
2/ Legend: OC = open circuit
0 = logical zero, digital type A signal (item 3.1 of table 1)
ON = 0.4 ± 0.4 Vdc
OFF = 13.5 ± 2.0 Vdc

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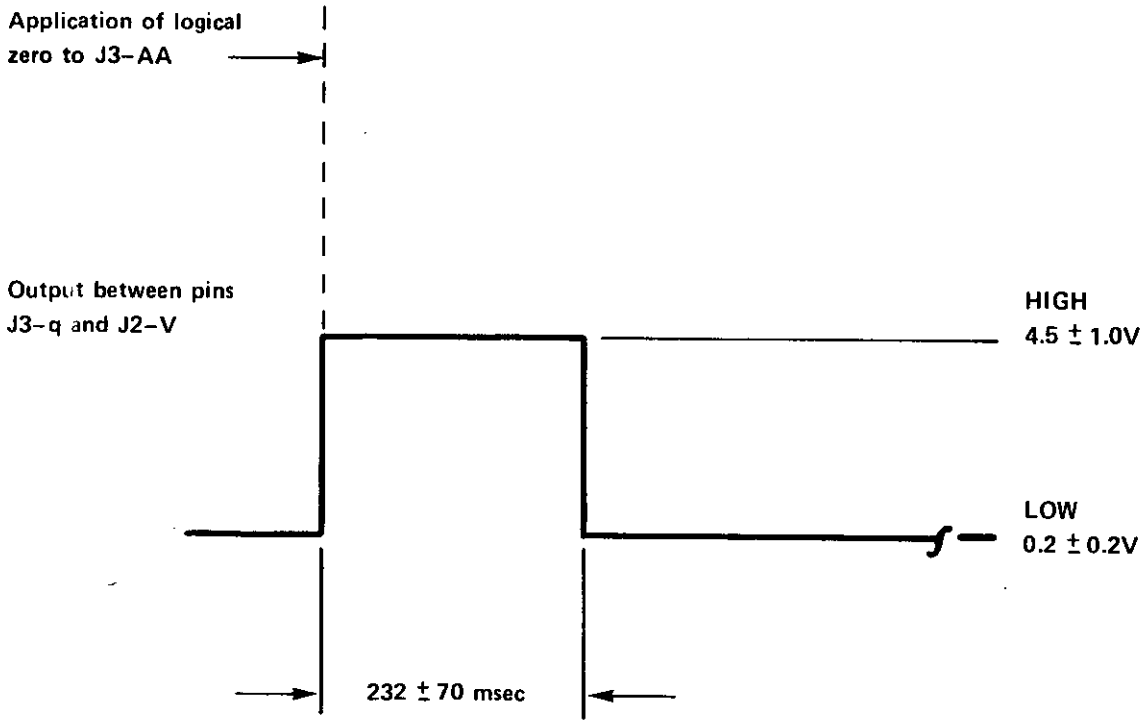


FIGURE 4. Blanking output waveform.

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TABLE VII. Test access outputs.

Item	Inputs at Pins of J2 1/ 2/ 3/						Outputs of Pins J7 1/ 3/ 4/		
	m	n	j	k	c	d	G	F	B
1	0	1	NA	NA	NA	NA	H_1	NA	NA
2	1	0	NA	NA	NA	NA	L	NA	NA
3	NA	NA	0	1	NA	NA	NA	H_2	NA
4	NA	NA	1	0	NA	NA	NA	L	NA
5	NA	NA	NA	NA	0	1	NA	NA	H_1
6	NA	NA	NA	NA	1	0	NA	NA	L

1/ Input and output signal voltages referended to J2-V.
2/ Input signal legend: 1 = logical one, digital type - A
0 = logical zero, digital type - A
3/ NA = not applicable:
4/ Output signal legend: $H_1 = 4.5 \pm 1.2$ V
$H_2 = 3.6 \pm 1.4$ V
$L = 0.2 \pm 0.2$ V

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TABLE VIII. Target Select, $\overline{\text{RESET}}$, $\overline{\text{RBI}}$, $\overline{\text{FEED}}$, and $\overline{9995}$.

Item	Input Conditions at Pins of J3 1/, 2/							Outputs 1/, 3/				
	W 7/	AA 7/	Y	Z	a	U	V	J3-m	J3-n	J3-k	Returns 5/	Range 5/ ±15
1	0 4/	NA	NA	NA	NA	0	NA	OFF	OFF	ON	0	0000
2	OC	0 4/	NA	NA	NA	NA	NA	OFF	ON	ON	1	830
3	0	NA	0	NA	NA	NA	NA	OFF	ON	ON	1	830
4	0	NA	NA	0	NA	NA	NA	OFF	ON	ON	1	830
5	0	NA	NA	NA	0	NA	NA	OFF	ON	ON	1	830
6	OC	0 4/	NA	NA	NA	NA	NA	OFF	ON	ON	2	1830
7	0 4/	0 4/	NA	NA	NA	NA	NA	OFF	ON	ON	3	2830
8	0 4/	0 4/	NA	NA	NA	NA	NA	ON	OFF	ON	4	2830

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TABLE VIII. Target Select, $\overline{\text{RESET}}$, $\overline{\text{RBI}}$, $\overline{\text{FEED}}$, and $\overline{9995}$. - Continued

Item	Input Conditions at Pins of J3 1/, 2/							Outputs 1/, 3/				
	W 7/	AA 7/	Y	Z	a	U	V	J3-m	J3-n	J3-k	Returns 5/	Range 5/ ±15
9	OC	NA	0	NA	NA	NA	NA	ON	OFF	OFF	4	830
10	OC	NA	NA	0	NA	NA	NA	ON	OFF	OFF	4	1830
11	OC	NA	NA	NA	0	NA	NA	ON	OFF	ON	4	2830
12	OC	NA	NA	NA	NA	NA	0	OFF	ON	ON	4	2830
13	0	NA	NA	NA	NA	0	NA	OFF	OFF	ON	0	0000
14	OC	NA	NA	0	NA	NA	NA	OFF	OFF	OFF	0	0000
15	OC	0 4/	NA	NA	NA	NA	NA	ON	OFF	OFF	1	9995 6/

1/ Input and output signal voltage referenced to Pin J2-V.
2/ Input signal legend: OC = open circuit
0 = logical zero, digital type - A signal (item 3.1 of table I)
NA = Not applicable
3/ Output legend: ON = 0.4 ± 0.4 Vdc
NA = Not applicable
OFF = 13.5 ± 2.0 Vdc
4/ Signal applied for not less than one second and not greater than two seconds.
5/ Readouts to the CLRC.
6/ Range readout may be 000 or 9990.
7/ Application of signal to J3-W shall precede signal to J3-AA.

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3.3.17 BIT test. With the connected as specified in, J2-h connected to J2-j, J2-k connected to J2-i, and with a logical zero, digital type A signal applied between J6-C and J6-L, return, the output between J6-F and J6-L shall be 0.2 ± 0.2 Vdc, and the shall indicate 1830 ± 15. (See 4.6.2.17)

3.3.18 Laser analog range. With the power sources (item 2.3, 2.4 and 2.5 of table I) connected to the specified pins and with a logical zero digital type - A signal (item 3.1 of table I) applied for not less than one second and not greater than two seconds between J3-U and J3-M and continuously between J3-R and J3-M and with the Counters (A5) Circuit Card Assembly removed from the unit and the lines from pins 4A and 6A of the Switch and Light Control (A10) assembly open to the pins of XA10 connector, application of a logical zero digital type A signal (item 3.1 of table I) to the pins specified in table IX, in any order, followed by application for not less than one second and not greater than two seconds of a logical zero digital type A signal between J3-X and J3-M shall cause the corresponding item outputs between J6-D and J6-L to be as specified in table IX. (See 4.6.2.18)

3.3.19 PFN low charge. With the assembly LVPS emergency power switch S1 in the XMTR test position, the output between J2-x and J2-V shall be 6.9 ± 0.7 V. (See 4.6.2.19)

3.3.20 MINIMUM RANGE INHIBIT 2. With J2-h connected to J2-j and J2-k connected to J2-i and with a logical zero digital type - A signal (item 3.1 of table I) applied between J3-R and J2-V and switch S1 on the unit Reply Gating Circuit Card Assembly 10559295 not depressed, the output waveform between J2-w and J2-V shall be as specified on figure 5. (See 4.6.2.20)

3.3.21 MALF 3 signal. With the unit connected and with a logical zero digital type - A signal (item 3.1 of table I) applied between J2-f and J2-V, the output shall indicate 0003. (See 4.6.2.21)

3.3.22 Input transient. The unit shall be capable of operation as specified herein when subjected to tank power input voltage transients from 9 volts to 90 volts with durations up to 100 milliseconds. (See 4.6.2.22)

3.3.23 PFN discharge time. With the computer on power removed (item 2.2, table I), the PFN current between pins A (+) and N (-) of test access connector J7 shall decrease to less than 7 microamperes within 25 seconds. (See 4.6.2.23)

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3.4 Sealing. With an internal nitrogen pressure of 5.0 ± 0.1 psig, the unit shall exhibit a pressure drop not greater than 0.2 psig in not less than 2 hours. The pressurized unit shall be purged with dry nitrogen in accordance with BB-N-411, type I, grade B, class 1, with a dewpoint of -25°F or lower and shall have a final pressure of 0.0 + 0.2 - 0.0 psig. (See 4.6.4)

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TABLE IX. Laser analog range outputs.

	Input Conditions A1XA5 Pins 1/													Output (J6-D)	
Item	1	2	3	4	5	6	7	8	9	10	11	12	13	Voltage 2/ (millivolts)	Ratio 3/
1	+	+	+	+	+	+	+	+	+	+	+	+	+	less than 4	--
2	+	+	+	+	+	+	G	+	+	+	+	+	G	218.1 ± 4	0.04362 ± 0.00080
3	+	+	+	+	+	+	G	+	+	+	+	G	+	223.4 ± 4	0.04468 ± 0.00080
4	+	+	+	+	+	+	G	+	+	+	G	+	+	234.1 ± 4	0.04682 ± 0.00080
5	+	+	+	+	+	+	G	+	+	G	+	+	+	255.4 ± 4	0.05108 ± 0.00080
6	+	+	+	+	+	+	G	+	G	+	+	+	+	297.9 ± 4	0.05958 ± 0.00080
7	+	+	+	+	+	+	G	G	+	+	+	+	+	319.2 ± 4	0.06384 ± 0.00080

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TABLE IX. Laser analog range outputs. - Continued

Item	Input Conditions A1XA5 Pins 1/													Output (J6-D)	
	1	2	3	4	5	6	7	8	9	10	11	12	13	Voltage 2/ (millivolts)	Ratio 3/
8	+	+	+	+	+	G	+	+	+	+	+	+	+	425.6 ± 4	0.08512 ± 0.00080
9	+	+	+	+	G	+	+	+	+	+	+	+	+	851.2 ± 4	0.17024 ± 0.00080
10	+	+	+	G	+	+	+	+	+	+	+	+	+	1064.0 ± 4	0.21280 ± 0.00080
11	+	+	G	+	+	+	+	+	+	+	+	+	+	2128.0 ± 6	0.42560 ± 0.00120
12	+	G	+	+	+	+	+	+	+	+	+	+	+	4256.0 ± 8	0.85120 ± 0.00160
13	G	G	G	G	G	G	G	G	G	G	G	G	G	1700 ± 250	0.34 ± 0.05

1/ Input signal legend
G = Pin grounded to J6-L
+ = Pin connected to 5 Vdc through individual 2- kilohm resistors (item 1.10 of table 1).

2/ Output voltage in rms (AC Ref) or dc (DC Ref)

3/ Ratio: $\frac{\text{Output voltage at J6-D (Vac)}}{\text{Ref voltage at J6-M (Vac)}}$ or $\frac{\text{Vdc}}{\text{Vdc}}$

When this method is used to verify the output, the tolerance on the Ref. voltage (item 2.5 of table 1) may be ± 1%.

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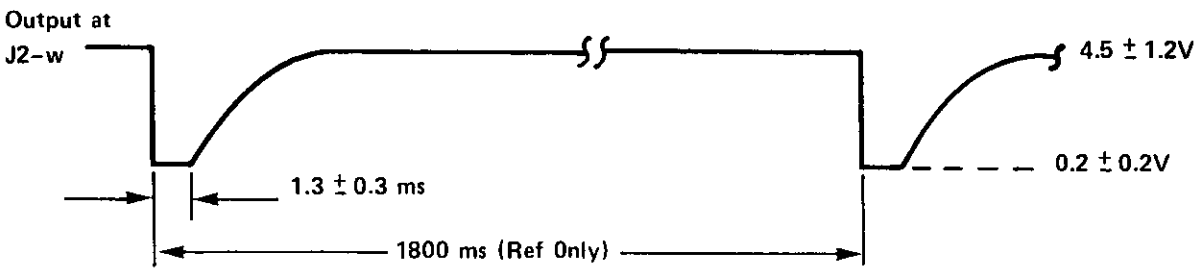


FIGURE 5. Minimum range inhibit (2) output waveform.

3.5 Environmental.

3.5.1 Shock. The unit shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the unit shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.5.2 Vibration. The unit shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 6. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.5.4 Operating temperature. The unit shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.5.4 Storage temperature. The unit shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5).

3.5.5 Humidity. The unit shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.3.6)

3.5.6 Altitude. The unit shall operate as specified herein under the following conditions:

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a. during exposure to an atmospheric pressure equivalent to 10000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (See 4.6.3.7)

3.5.7 Dust. The unit shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (See 4.6.3.8)

3.5.8 Salt fog. The unit shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.3.9)

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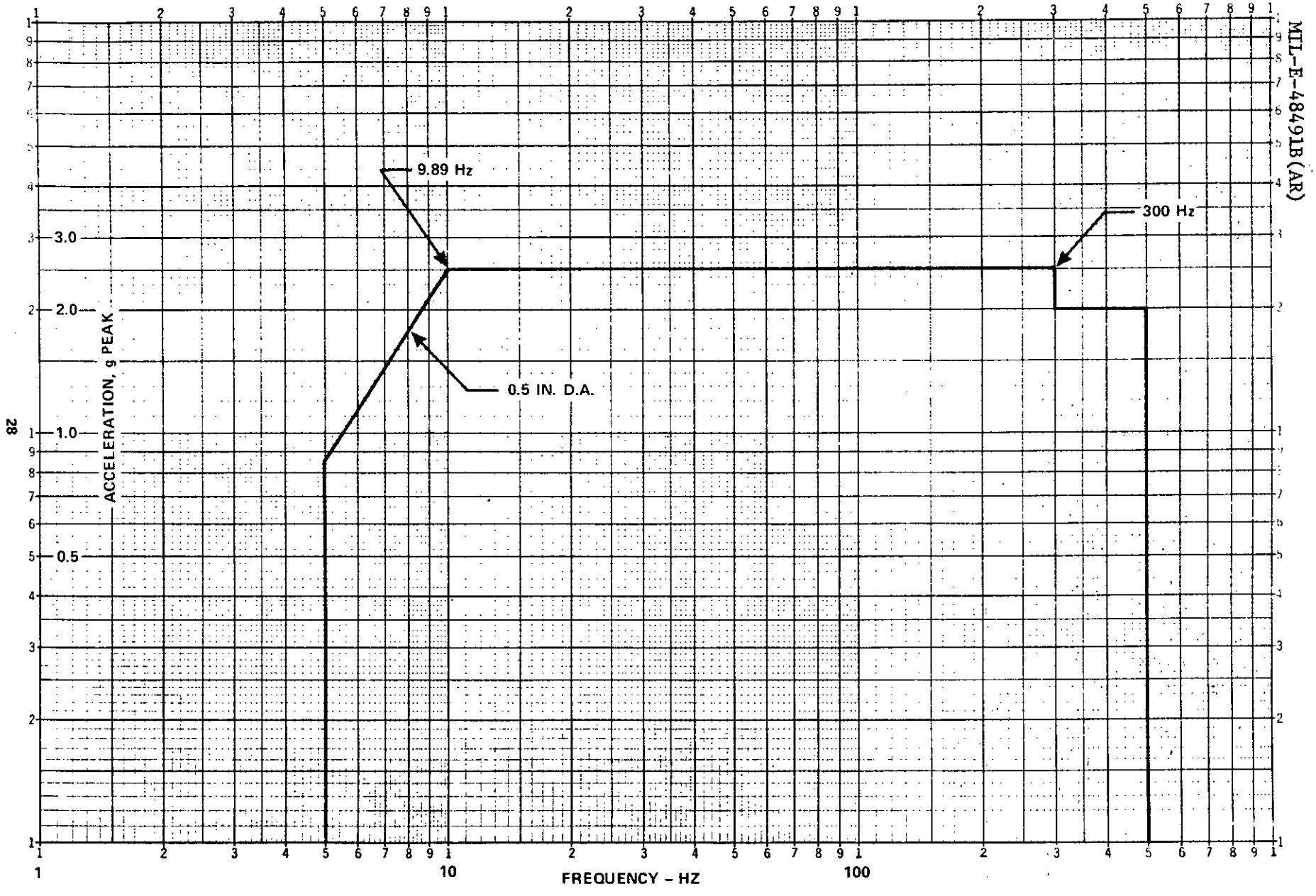


FIGURE 6. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three units selected at random by the Government representative from the first fifteen production units.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in table X.

4.4.3 Failure. Failure of any unit to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any unit to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE X. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Fabrication	3.1	Applicable drawing - Visual
2	General Specifications	3.1.2	MIL-F-13926 - Visual
3	Sealing A and B	3.4	4.6.4
4	Performance tests	3.3	4.6.2
5	Low temperature	3.5.3 and 3.5.4	4.6.3.4
6	High temperature	3.5.3 and 3.5.4	4.6.3.3
7	Altitude	3.5.6	4.6.3.7
8	Dust	3.5.7	4.6.3.8
9	Shock	3.5.1	4.6.3.1
10	Vibration A	3.5.2	4.6.3.2
11	Humidity	3.5.5	4.6.3.6
12	Salt fog	3.5.8	4.6.3.9
13	Temperature cycling	3.5.3 and 3.5.4	4.6.3.5

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TABLE XI. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Circuit continuity and isolation	3.3.1	4.6.2.1
102	Power interlock	3.3.2	4.6.2.2
103	Remote power-on	3.3.3	4.6.2.3
104	Initial turn-on conditions and battle range	3.3.4	4.6.2.4
105	PFN current sensing	3.3.5	4.6.2.5
106	PFN voltage	3.3.6	4.6.2.6
107	400 V supply	3.3.7	4.6.2.7
108	$\overline{\text{FIRE}}$ and $\overline{\text{MALF 4}}$ signals	3.3.8	4.6.2.8
109	$\overline{\text{MALF 2}}$ signal	3.3.9	4.6.2.9
110	Range readout	3.3.10	4.6.2.10
111	Switch and light control	3.3.11	4.6.2.11
112	Blanking	3.3.12	4.6.2.12
113	$\overline{\text{MANUAL RESET}}$	3.3.13	4.6.2.13
114	Test access outputs	3.3.14	4.6.2.14
115	Target select, $\overline{\text{RESET}}$, $\overline{\text{RBI}}$, $\overline{\text{FEED}}$ and $\overline{9995}$	3.3.15	4.6.2.15
116	Battle range logic	3.3.16	4.6.2.16
117	$\overline{\text{BIT}}$ test	3.3.17	4.6.2.17

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TABLE XI. Performance tests. - Continued

Item	Characteristic	Requirement	Test Procedure
118	Laser analog range	3.3.18	4.6.2.18
119	PFN Low charge	3.3.19	4.6.2.19
120	$\overline{\text{MINIMUM RANGE INHIBIT}}$ (2)	3.3.20	4.6.2.20
121	$\overline{\text{MALF}}$ 3 signal	3.3.21	4.6.2.21
122	Input transient	3.3.22	4.6.2.22
123	PFN discharge time	3.3.23	4.6.2.23
124	Sealing A	3.4	4.6.4

TABLE XII. General tests.

Item	Characteristics	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.1.2	MIL-F-13926 - Visual

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TABLE XIII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.5.1	4.6.3.1
304	Vibration B	3.5.2	4.6.3.2
305	High Temperature	3.5.3 and 3.5.4	4.6.3.3
306	Low Temperature	3.5.3 and 3.5.4	4.6.3.4
307	Temperature cycling	3.5.3 and 3.5.4	4.6.3.5
308	Sealing B	3.4	4.6.4

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4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the unit is procured for use as a:

Use	Plan
a. Logistics spare unit apart from the next higher assembly	A and C
b. Component unit of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One unit, as a control sample, shall be selected at random by the Government representative from each 100 units produced and shall be subjected to all the tests in table XII.

4.5.2.2 Environmental sample and tests. Three units, as a control sample, shall be selected at random by the Government representative from each 50 units produced or from each month's production, whichever occurs first. All the tests in table XIII shall be applied separately to each unit in the sample.

4.5.2.3 Acceptance. Where any one unit of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table XII and only the test of item 307 in table XIII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each unit in every lot shall be subjected to all the tests in table XI and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one unit fails to meet any specified requirement, the defective unit shall be removed from the lot and resubmitted only after all defects have been corrected.

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4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11743131.

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table XIV shall be used where specified to perform the required test.

TABLE XIV. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750439	Adapter - Electronic Unit
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
5	11821471	Adapter - Handling Fixture, Altitude
6	11821520	Automatic Test Assembly Central Processor Unit
7	11821748	Immersion Leak, Purge and Fill

4.6.2 Performance tests.

4.6.2.1 Circuit continuity and isolation. Use items 1, 2 and 6 in table XIV to verify compliance with the requirements of 3.3.1.

4.6.2.2 Power interlock. Use items 1, 2 and 6 in table XIV to verify compliance with the requirements of 3.3.2.

4.6.2.3 Remote power-on. Use items 1, 2 and 6 in table XIV to verify compliance with the requirements of 3.3.3.

4.6.2.4 Initial turn-on conditions and battle range. Use items 1, 2 and 6 in table XIV to verify compliance with the requirements of 3.3.4.

4.6.2.5 PFN current sensing. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.5.

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4.6.2.6 PFN voltage adjustment and programming. Use items 1, 2 and 6 in table XIV to verify compliance with the requirements of 3.3.6.

4.6.2.7 Four hundred Vdc supply. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.7.

4.6.2.8 $\overline{\text{Fire}}$ and $\overline{\text{MALF 4}}$ signals. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.8.

4.6.2.9 $\overline{\text{MALF 2}}$ signal. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.9.

4.6.2.10 Range readout. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.10.

4.6.2.11 Switch and light control. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.11.

4.6.2.12 Blanking. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.12.

4.6.2.13 $\overline{\text{MANUAL RESET}}$. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.13.

4.6.2.14 Test access outputs. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.14.

4.6.2.15 Target select, $\overline{\text{RESET}}$, $\overline{\text{RBI}}$ (ripple blanking input), feed, and 9995. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.15.

4.6.2.16 Battle range logic. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.16.

4.6.2.17 $\overline{\text{BIT}}$ test. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.17.

4.6.2.18 Laser analog range. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.18.

4.6.2.19 PFN low charge. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.19.

4.6.2.20 $\overline{\text{MINIMUM RANGE INHIBIT 2.}}$ Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.20.

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4.6.2.21 MALF 3 signal. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.21.

4.6.2.22 Input transient. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.22.

4.6.23 PFN discharge time. Use items 1, 2 and 6 of table XIV to verify compliance with the requirements of 3.3.23.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3 and 4 of table XIV and subject the unit to the shock test specified in MIL-STD-810, method 516.2, procedure I figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ± 4 g with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the unit to the performance tests of 4.6.2.

4.6.3.1.2 High intensity. Use items 3 and 4 of table XIV and subject the unit to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ± 10.0 g with a time duration of 1.5 ± 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the unit to the performance tests of 4.6.2.

4.6.3.2 Vibration. Application of Vibration A and Vibration B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.6.3.2.1 Vibration A. Use item 4 of table XIV and subject the unit to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve shown in figure 6 and duration of exposure shall be not less than 80 miniutes per axis. At the conclusion of this test, subject the unit to the performance tests of 4.6.2.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shll be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Subject the unit to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of 4.6.2 shall be applied.

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4.6.3.4 Low temperature. Subject the unit to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of 4.6.2 shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the unit to the temperature profile specified in figure 7. At the conclusion of this test, subject the unit to the performance tests of 4.6.2.

4.6.3.6 Humidity. Place the unit in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the unit to the performance tests of 4.6.2. After exposure, subject the unit to the performance tests of 4.6.2.

4.6.3.7 Altitude. Use item 5 of table XIV, place the unit in an altitude chamber and subject it to the altitude test specified in MIL-STD-810, method 500.1, procedure I, except that:

a. First, the unit in a non-operating conditions, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the unit to the performance tests of 4.6.2.

4.6.3.8 Dust. Subject the unit to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, subject the unit to the performance tests of 4.6.2.

4.6.3.9 Salt fog. Subject the unit to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this test, subject the unit to the performance tests of 4.6.2.

4.6.4 Sealing. Pressurize the unit with dry nitrogen to 5.0 ± 0.1 psig and perform either Sealing A or Sealing B leakage test. After determination of compliance with the sealing requirement, purge the unit with dry nitrogen at 1 psig for not less than 5 minutes and lower the pressure to 0.0 + 0.2 - psig. Nitrogen shall be BB-N-411, type I, class 1, grade B with a dewpoint not greater than -25°F.

4.6.4.1 Test selection. Application of Sealing A and Sealing B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

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4.6.4.2 Sealing A. Use item 7 of table XIV and subject the unit to the leakage (immersion) test specified in MIL-STD-810, method 512.1, procedure IV. The immersion period shall be not less than 5 minutes.

4.6.4.3 Sealing B. Use item 7 of table XIV, place the unit in the following environment for not less than 2 hours and verify that the pressure drop is not greater than 0.2 psig.

a.	Temperature	73° ± 2.5°F
b.	Relative humidity	50 percent ± 5 percent
c.	Atmospheric pressure	28.5 + 2.0 − 3.0 in. Hg.

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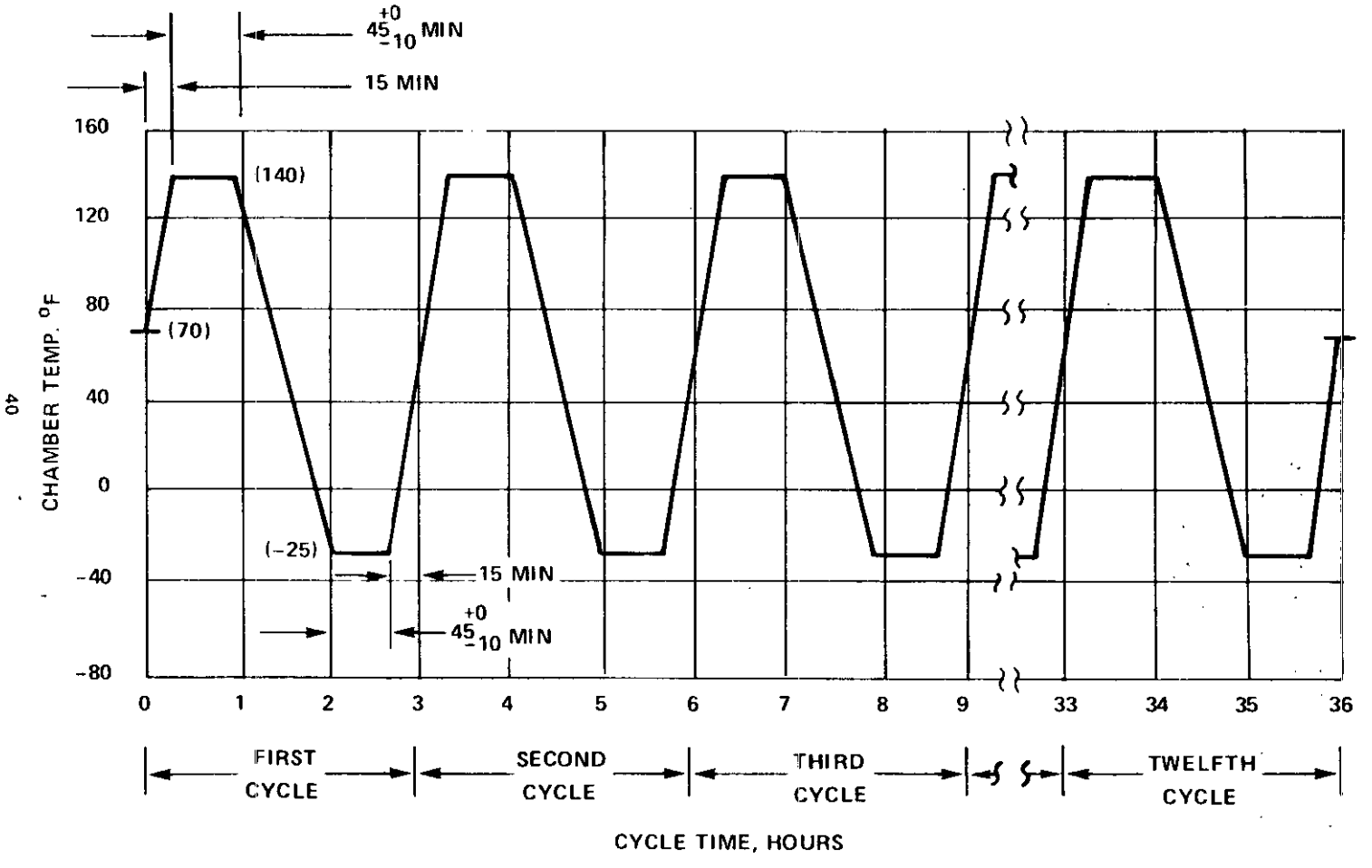


FIGURE 7. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11743131. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Electronic Unit is a unit of the Laser Fire Control Rangefinder: AN/VVG-2. The unit provides system power supply and control, laser pulse forming, reply gating and selection, range counting, computer input line driving, malfunction detection and emergency power control.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Special tests and examinations as required, the type of inspection required and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-E-48491B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-48491B (AR)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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